



Case report

Primary spinal epidural hematoma mimicking stroke: A case report of rare diagnostic challenge

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ABSTRACT

Introduction and importance: Spinal epidural hematoma is a rare and potentially life-threatening condition characterized by bleeding into epidural space, leading to spinal cord compression. With an incidence of approximately 1 per 1,000,000 annually, SEH requires urgent diagnosis and management to prevent irreversible neurological damage. This report highlights a case of primary spinal epidural hematoma causing acute spinal cord compression and hemiplegia mimicking stroke like features.

Case presentation: A 23-year-old male presented to the emergency department with acute upper back pain, progressive lower limb weakness, and bowel and bladder incontinence. Neurological examination revealed motor strength of 0/5 in the lower limbs, sensory loss in the perineal region, and preserved bulbocavernosus reflex. MRI of the spine demonstrated epidural hematoma compressing the spinal cord from C6 to D2. The patient underwent emergent laminotomy and hematoma evacuation. Postoperative recovery was uneventful, with significant improvement in neurological function.

Clinical discussion: Spinal epidural hematoma is a diagnostic challenge due to its nonspecific symptoms, which may mimic other conditions such as ischemic stroke. MRI remains the diagnostic gold standard. Timely surgical decompression is critical for favorable outcomes, especially in cases presenting significant neurological deficits.

Conclusion: This case emphasizes the importance of rapid recognition of SHE from stroke and spinal cord ischemia, imaging, and surgical management of Spinal epidural hematoma to mitigate severe neurological consequences. Awareness of this rare condition is vital for early diagnosis and effective treatment.

1. Introduction

Primary Spinal epidural hematoma (SEH) is a rare condition characterized by bleeding in the epidural space, potentially causing spinal cord compression or cauda equina syndrome [1,2]. With an incidence of approximately 1 per 1,000,000 patients annually, Primary SEH typically presents with acute, severe pain at the hemorrhage site, often radiating to the extremities, and can rapidly progress to severe neurologic deficits [3]. The pathophysiology involves the accumulation of blood in the epidural space, leading to mechanical compression of the spinal cord, which can result in severe neurological deficits, including paraplegia or quadriplegia [4]. The clinical presentation of SEH varies widely, ranging from localized back pain to acute neurological deficits, which may

mimic other conditions such as ischemic stroke and spinal cord lesion such as transverse myelitis and spinal cord ischemia [5]. The urgency of diagnosis and intervention is critical, as delayed treatment can lead to irreversible neurological damage [6]. Surgical decompression remains the standard treatment for symptomatic cases, particularly when neurological deficits are present [7]. From a modern perspective, the challenge of spinal epidural hematoma does not lie in the ability to diagnose or treat the condition, but rather in diagnosing the underlying cause and preventing recurrence.

Here in, we will discuss a case of a 23-year-old male, presented with acute pain on the interscapular area, hemiplegia along with bowel and bladder incontinence which mimicked like ischemic stroke. This is reported according to SCARE guideline [8].

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2. Case presentation

A 23-year-old male presented to the emergency department with a chief complaint of acute pain in the upper back region, progressive weakness in both lower limbs, and bowel and bladder incontinence. The symptoms began two days prior with non-radiating pain localized to the interscapular area, followed by weakness in both lower limbs that rendered the patient unable to stand. He reported sensory loss in the perineal region and concurrent bowel and bladder incontinence. The patient denied any history of fever, nausea, vomiting, abdominal pain, heavy lifting, trauma, autoimmune disease in the family, recent vaccinations, or chronic illness. There was no prior use of medications. On physical examination, the patient was alert and oriented with a Glasgow

Coma Scale score of 15/15. Vital signs were stable. There were no signs of pallor, icterus, cyanosis, lymphadenopathy, or dehydration. Neurological examination revealed absent perianal sensation, although the bulbocavernosus reflex was preserved. Motor strength in the lower limbs was 0/5, while upper limb strength was 5/5.

Initial blood investigations showed elevated total leukocyte count (13,000/cmm; normal: 4000–11,000), neutrophilia (85 %; normal: 45–75 %), and slightly decreased lymphocyte count (10 %; normal: 24–45 %). Other hematological and biochemical parameters, including hemoglobin, renal function, and differential leukocyte count, were within normal limits. Coagulation studies performed on the patient revealed normal results. The prothrombin time (PT) was 12 s (normal range: 10–14 s), the international normalized ratio (INR) was 1.0

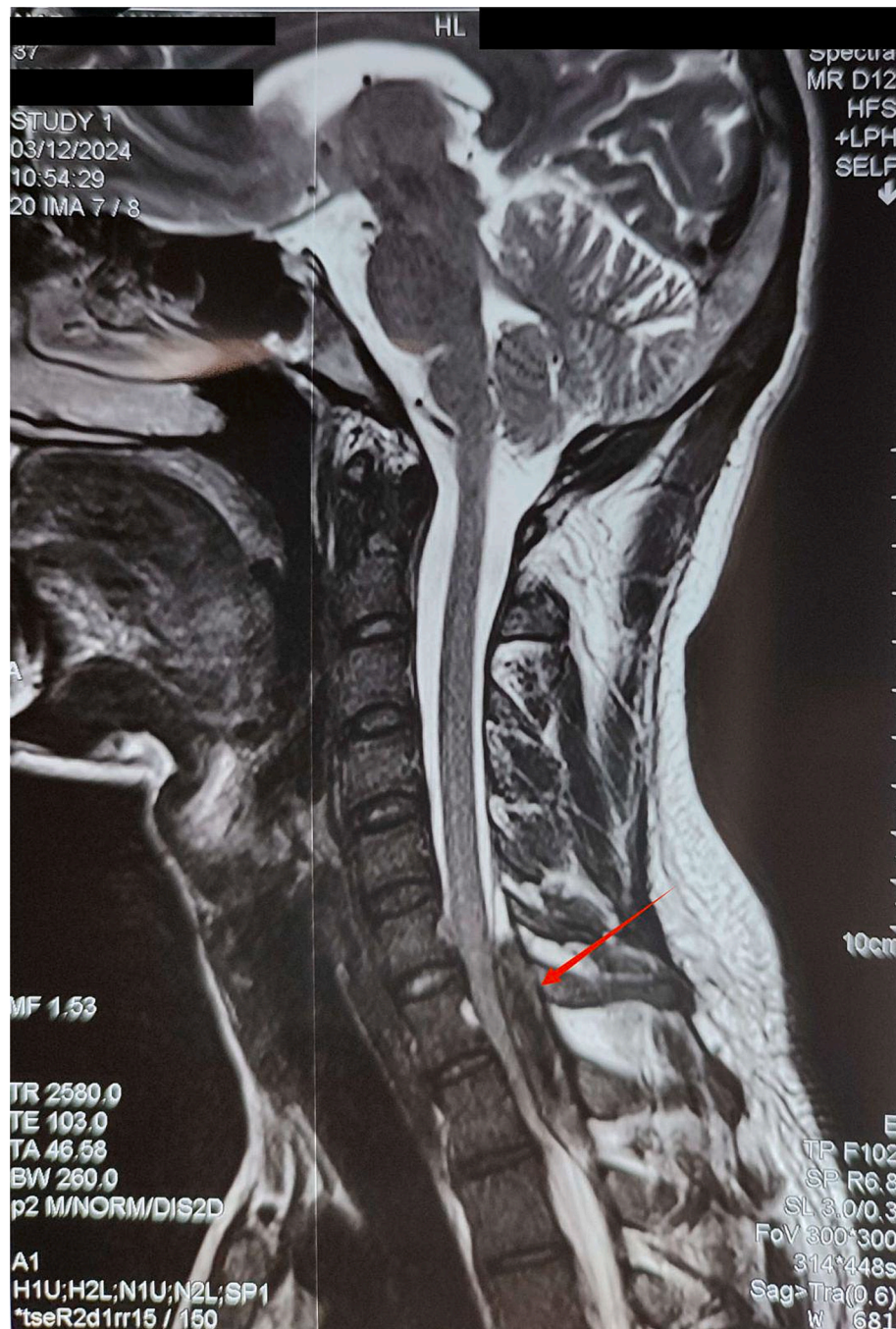


Fig. 1. Sagittal Magnetic Resonance Imaging (MRI) demonstrated an epidural hematoma compressing the spinal cord and posterior thecal sac from the C6 to D2 vertebral levels.

(normal range: 0.8–1.2), and the activated partial thromboplastin time (aPTT) was 30 s (normal range: 25–35 s). Fibrinogen levels were measured at 300 mg/dL (normal range: 200–400 mg/dL), and the platelet count was 250,000/ μ L (normal range: 150,000–400,000/ μ L). These findings ruled out any underlying coagulation disturbances that might have contributed to the development of spinal epidural hematoma. Chest X ray was performed and was normal. Magnetic resonance imaging (MRI) of the cervical spine and brainstem was performed. The brain and brainstem MRI was normal excluding stroke as an etiology. Sagittal and axial T2-weighted imaging demonstrated an epidural hematoma compressing the spinal cord and posterior thecal sac from the C6 to D2 vertebral levels (Figs. 1 and 2). There was no radiological evidence of trauma or structural abnormalities of spine as well as brain.

The diagnosis of spontaneous spinal epidural hematoma (SSEH) causing acute spinal cord compression was confirmed based on clinical presentation, laboratory findings, and MRI. The patient was promptly referred to the neurosurgery department for emergency surgical intervention. Following preoperative evaluations, the patient underwent laminotomy and hematoma evacuation under general anesthesia. A midline incision was made from C5 to D2, followed by paraspinal muscle dissection to expose the lamina. Hemi-laminotomy was performed, and the hematoma was evacuated. Hemostasis was achieved, and the incision was closed in layers (Figs. 3 and 4). During the laminotomy and hematoma evacuation, meticulous hemostasis was achieved using techniques such as bipolar cautery and absorbable hemostatic agents. After the evacuation of the hematoma, the surgical site was carefully inspected to identify and manage any active bleeding vessels or areas prone to rebleeding. The dura was examined to ensure there were no additional sources of bleeding. The incision was closed in layers to maintain tissue integrity and provide adequate support to the surgical site. This prevents undue strain on the wound, which could lead to reopening or minor bleeding. Postoperatively, the patient was closely monitored for signs of recurrence, such as worsening neurological deficits or new onset of back pain. In the immediate postoperative period, anticoagulants and antiplatelet agents were avoided unless clinically indicated to minimize the risk of rebleeding. Postoperative recovery was uneventful, with gradual improvement in neurological function. The patient was discharged after 2 days of hospital stay with instructions for follow-up and rehabilitation. The patient was followed up at 4, and 8

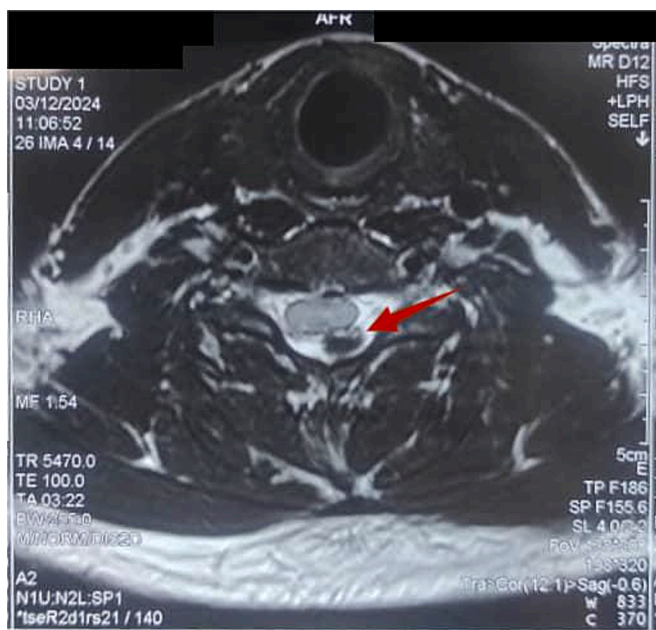


Fig. 2. Axial T2-weighted MRI demonstrated an epidural hematoma compressing the spinal cord.

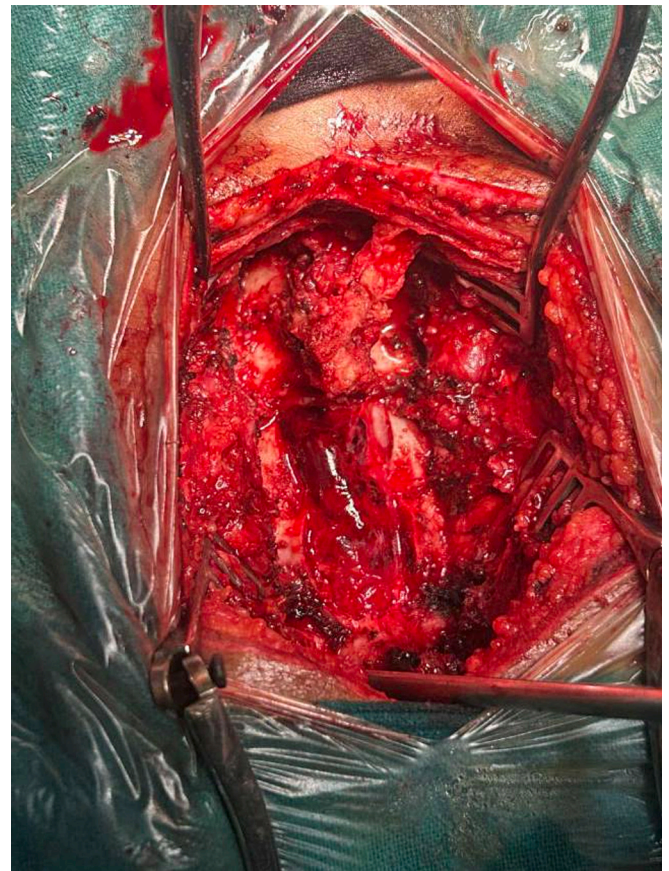


Fig. 3. Intraoperative procedure: Hemi Laminotomy performed with midline incision at C5 to D2, followed by paraspinal muscle dissection to expose the lamina, showing epidural hematoma.

weeks and the patient were fully recovered without any neurological deficit.

3. Discussion

The case presented illustrates the complexities surrounding the diagnosis and management of epidural hematoma resulting in spinal cord compression. The mechanism in such cases often involves the sudden accumulation of blood in the epidural space, which exerts pressure on the spinal cord. This pressure can lead to both immediate and secondary injuries to neural tissues, including damage to the vascular endothelium and surrounding soft tissues. These tissues may include the spinal cord, dura mater, blood vessels, ligaments, and other components of the epidural space. The pressure exerted by the hematoma on these tissues can lead to neural and vascular damage, influencing both immediate symptoms and long-term recovery outcomes [9]. The impact of epidural hematoma on spinal cord structures is profound. The immediate mechanical compression of the spinal cord due to the hematoma can lead to both primary and secondary injury mechanisms. Primary injury occurs due to direct compression of neural tissue, which can disrupt neuronal function and lead to immediate neurological deficits. Secondary injury mechanisms are also critical; they include ischemia resulting from compromised blood flow, inflammation, and the release of neurotoxic substances, which can exacerbate neuronal damage [1,19]. In this case, the rapid onset of neurological deficits necessitated prompt imaging and surgical intervention to alleviate the compression. Spinal cord hematomas are rare conditions classified based on the location of bleeding: intramedullary, subarachnoid, subdural, or epidural. Common causes include trauma, iatrogenic factors,



Fig. 4. Intraoperative procedure: Hemi Laminotomy, showing epidural hematoma after midline incision.

vascular malformations, and bleeding disorders and symptoms typically involve acute onset of pain and myelopathy, though presentation can vary. In our case, the patient developed acute pain in the interscapular area, hemiplegia along with bowel and bladder incontinence without any identifying cause. While often associated with trauma, anticoagulation therapy, or spinal procedures, primary SEH can occur without an obvious cause [11]. Diagnosis is challenging due to overlapping symptoms with ischemic stroke, but maintaining high clinical suspicion and using MRI can aid in accurate identification [10]. The misdiagnosis of SEH as a stroke can have serious implications for treatment, especially regarding the administration of thrombolytics like tissue plasminogen activator (TPA), which could exacerbate the hematoma and complicate surgical intervention [17]. Diagnostic imaging, particularly magnetic resonance imaging (MRI), plays a crucial role in the identification of SEH. MRI can effectively delineate the extent of the hematoma and its impact on surrounding neural structures, aiding in the differentiation from other conditions such as stroke [18]. In our case, MRI delineated the confusion between hematoma and ischemic stroke in brain.

The literature indicates that the timing of surgical intervention is crucial for optimal recovery outcomes. Studies have shown that surgical decompression performed within 12 h of symptom onset significantly improves neurological recovery [12]. In contrast, delays in treatment can result in permanent deficits, as evidenced by cases where patients are presented with extensive neurological impairments due to prolonged compression [4,13]. In our case, the patient underwent surgical intervention within 2 days of symptoms however no significant neurological impairments were noticed on the follow up. Furthermore, the differential diagnosis for patients presenting with back pain and neurological deficits must include EH, as misdiagnosis can lead to inappropriate

management strategies [6,14]. Interestingly, while spontaneous resolution of epidural hematomas has been documented, this phenomenon is rare and typically occurs in cases with minimal initial symptoms [13]. In our case, the decision to proceed with surgical intervention was supported by the presence of significant neurological deficits, aligning with the recommendations in the literature that advocate for surgical management in symptomatic patients [4,6]. Preoperative neurological status significantly influences outcomes, with incomplete deficits correlating to better recovery and shorter hospital stays. Factors contributing to poor postoperative recovery include coagulopathy, larger hematoma size, and complete preoperative spinal dysfunction [15].

While surgery is generally recommended, conservative management may be considered for patients without neurological deficits [16]. In this case, due to neurological deficit, urgent surgery was recommended after diagnosis through history, examination and radiology. This case highlights several important clinical insights. First, it emphasizes the need to consider SEH in the differential diagnosis for acute neurological deficits resembling ischemic stroke, particularly in young patients without traditional risk factors such as trauma or anticoagulation therapy. Misdiagnosis of SEH as stroke can lead to inappropriate management, including the administration of thrombolytics, which may worsen the hematoma and complicate subsequent surgical intervention. Second, the case underscores the critical role of MRI in differentiating SEH from other neurological conditions, facilitating timely and accurate diagnosis. Third, it illustrates the importance of early surgical decompression, which is pivotal in achieving favorable neurological outcomes. Finally, this case serves as a valuable teaching tool for emergency and neurology teams, reinforcing the importance of maintaining high clinical suspicion for rare conditions like SEH, even in the absence of apparent predisposing factors.

4. Conclusion

Primary Spinal epidural hematoma is a rare but potentially severe condition requiring high clinical suspicion for timely diagnosis and management. MRI is the gold standard for identifying SEH, particularly in patients presenting with acute neurological deficits. Prompt surgical decompression, preferably within 12 h of symptom onset, significantly improves neurological recovery and reduces the risk of permanent deficits. Conservative management may be considered in asymptomatic cases, but surgical intervention is critical for cases with significant neurological compromise, as illustrated by the presented case. Factors such as coagulopathy, larger hematoma size, and complete preoperative spinal dysfunction are associated with poorer outcomes.

CRediT authorship contribution statement

Bibek Shrestha Conceptualization, data curation, formal analysis, methodology, project administration, original writing, review, editing, and visualization

Bishal Gaurav Supervision, validation, and investigations

Sushil K Shilpakar Supervision, validation, and investigations.

Dipendra Thapa Supervision, validation, and investigations.

Bikas Thapa Supervision, validation, and investigations.

Consent

Written informed consent was obtained from the patient for publication of this case report and accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal on request.

Guarantor

Dr. Bishal Gaurav

Provenance and peer review

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Declaration of competing interest

None.

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